Reservoir and spin dynamics of a polariton condensate

Korostelev Nikita

Aim of the study

Supplement to the report on exciton-polariton condensation.

The previous report, "Dynamics and spatial structure of exciton-polariton condensates under nonresonant pumping," considered a scalar polariton condensate coupled to a single reservoir. Here, the model is extended in successive steps by introducing an inactive reservoir, two circularly polarized condensate components, and coherent coupling between them.

Three questions are addressed: how the additional reservoir changes the transient dynamics; how pump polarization affects the densities of the spin components; and at which coupling strengths the densities and relative phase become constant in time. The main parameters of the previous model are retained. The choice of the additional coefficients is explained in Appendix A.

1 Active and inactive reservoirs

1.1 Extension of the scalar model

In the previous model, the condensate field ψ and reservoir density n_R obeyed

$$\begin{gathered} i\hbar\frac{\partial\psi}{\partial t} = \left[-K\nabla^2 + g_C|\psi|^2 + g_R(n_R+\eta P_A) + \frac{i\hbar}{2}(Rn_R-\gamma_C)\right]\psi, \\ \frac{\partial n_R}{\partial t} = P_A - (\gamma_R + R|\psi|^2)n_R, \qquad K=\frac{\hbar^2}{2m}. \end{gathered} \tag{1}$$

Here, P_A is the particle injection rate into the reservoir. This notation replaces the previous P to distinguish external pumping from the supply of particles to the active reservoir.

The term $g_R\eta P_A$ describes an additional energy shift that follows the pump instantaneously. This contribution is now represented by a separate dynamical density n_I. The inactive exciton reservoir is populated by the external pump P_{in} and supplies the active reservoir n_A at a rate Wn_I. The active reservoir directly supplies particles to the condensate through stimulated scattering:

$$\begin{gathered} i\hbar\frac{\partial\psi}{\partial t} = \left[-K\nabla^2 + g_C|\psi|^2 + g_An_A + g_In_I + \frac{i\hbar}{2}(Rn_A-\gamma_C)\right]\psi, \\ \frac{\partial n_A}{\partial t} = Wn_I - (\gamma_A + R|\psi|^2)n_A, \\ \frac{\partial n_I}{\partial t} = P_{\text{in}} - (\gamma_I + W)n_I. \end{gathered} \tag{2}$$

Both reservoirs shift the condensate energy. The previous term $g_R\eta P_A$ is not added again in Eq. (2). The scalar model and its extension with active and inactive reservoirs are based on Refs. [1, 2].

To match the models, we choose

$$g_A = g_R, \qquad \gamma_A = \gamma_R, \qquad P_{\text{in}} = \frac{\gamma_I + W}{W}P_A, \qquad \frac{g_I}{W} = g_R\eta. \tag{3}$$

Under constant pumping, $n_I^* = P_A/W$. The supply of particles to the active reservoir and the additional energy shift therefore reproduce their previous values:

$$Wn_I^* = P_A, \qquad g_In_I^* = g_R\eta P_A.$$

The stationary solutions of the scalar model are thus preserved. In the spatially homogeneous case, the condensation threshold is also unchanged:

$$P_{\mathrm{th}} = \frac{\gamma_C \gamma_A}{R} = 0.43333\ \mu\mathrm{m}^{-2}\,\mathrm{ps}^{-1}.$$

This matching does not require the transient dynamics to coincide.

1.2 Response to pump turn-on and turn-off

Define the particle transfer rate into the active reservoir as

$$Q_A = W n_I.$$

Equations (2) and the matching conditions (3) give

$$\tau_I \frac{dQ_A}{dt} = P_A - Q_A, \qquad \tau_I = \frac{1}{\gamma_I + W}.$$

The supply to the active reservoir now responds on a finite time scale τ_I, rather than changing instantaneously. For the main parameter set, $\tau_I = 9.63$ ps.

The comparison uses $P_A = 1.5 P_{\mathrm{th}}$, initially empty reservoirs, and the same weak condensate seed. Both models approach the density

$$n_C^* = \frac{P_A - P_{\mathrm{th}}}{\gamma_C} = 1.19048\ \mu\mathrm{m}^{-2}.$$

Condensate growth starts later in the model with two reservoirs. Half of the steady-state density is first reached at 159.1 ps in the previous model and at 188.0 ps in the extended model. This delay reflects the entire transient and is not equal to τ_I.

After pump turn-off, the inactive reservoir continues to supply particles to the active reservoir for some time. In the previous model, the external supply to the active reservoir stops immediately, although particles already stored in that reservoir can still scatter into the condensate.

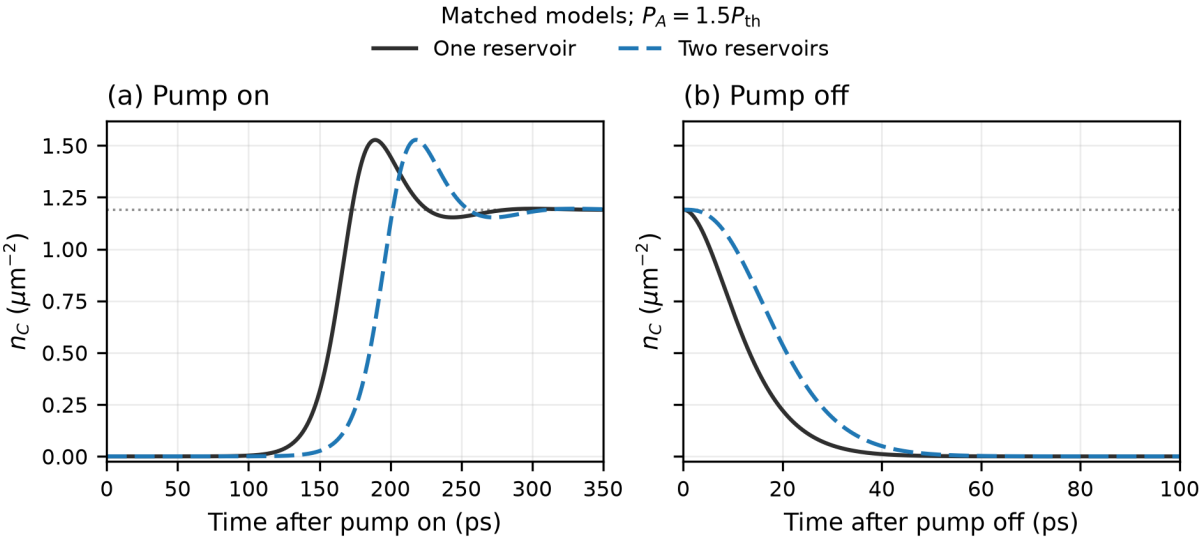


Figure 1: Homogeneous condensate density $n_C = |\psi|^2$ in the one- and two-reservoir models. **(a)** Growth after pump turn-on at $P_A = 1.5P_{\mathrm{th}}$; the horizontal dotted line marks the common steady density $n_C^* = 1.19048\ \mu\mathrm{m}^{-2}$. **(b)** Decay after turn-off at $t_{\mathrm{off}} = 1000$ ps, with time measured from turn-off. Both runs start with $n_C(0) = 10^{-4}\ \mu\mathrm{m}^{-2}$ and empty reservoirs. Parameters are listed in Appendix A.

2 Two spin components of the condensate

2.1 Model equations

In the circular polarization basis, the spinor polariton condensate is described by the fields ψ_+ and ψ_-. Their densities are $n_\pm = |\psi_\pm|^2$. Each component has an active and an inactive reservoir.

We use the coupled equations

$$\begin{aligned} i\hbar\frac{\partial\psi_\sigma}{\partial t} &= \left[-K\nabla^2 + V_\sigma + \frac{i\hbar}{2}(Rn_{A\sigma} - \gamma_C)\right]\psi_\sigma - J\psi_{-\sigma}, \\ V_\sigma &= \alpha_1 n_\sigma + \alpha_2 n_{-\sigma} + g_A n_{A\sigma} + g_I n_{I\sigma}, \\ \frac{\partial n_{A\sigma}}{\partial t} &= W n_{I\sigma} - (\gamma_A + Rn_\sigma)n_{A\sigma} + \Gamma_{s,A}(n_{A,-\sigma} - n_{A\sigma}), \\ \frac{\partial n_{I\sigma}}{\partial t} &= P_{\mathrm{in},\sigma} - (\gamma_I + W)n_{I\sigma} + \Gamma_{s,I}(n_{I,-\sigma} - n_{I\sigma}), \qquad \sigma = \pm. \end{aligned} \tag{4}$$

The coefficients α_1 and α_2 describe energy shifts due to interactions between polaritons with the same and opposite circular polarizations, respectively. The terms proportional to Γ_s describe reservoir spin relaxation: they transfer particles between the two spin components while conserving their total density. This spinor model structure is used in Ref. [3].

In this section, $J = 0$; coherent coupling is considered in Section 3. Each condensate component experiences the energy shift from the reservoir of the same circular polarization; cross-spin reservoir interaction coefficients are set to zero.

All interactions in Eq. (4) are local. Different spatial points are coupled through the kinetic operator $-K\nabla^2$. Reservoir diffusion is absent from this model.

2.2 Density dependence on pump polarization

For the homogeneous problem, we set

$$P_{A,\pm} = sP_{\mathrm{th}}(1 \pm \varepsilon), \qquad P_{\mathrm{in},\pm} = \frac{\gamma_I + W}{W}P_{A,\pm}. \tag{5}$$

Here, $P_{A,\pm}$ denotes a rescaled external pump parameter. The actual transfer rate into each active reservoir is $Q_{A,\pm} = Wn_{I,\pm}$, which generally differs from $P_{A,\pm}$ because of spin relaxation in the inactive reservoir.

The parameter ε describes the pump imbalance between the two components. Varying it leaves the total pump $P_{A,+} + P_{A,-} = 2sP_{\mathrm{th}}$ unchanged. Below, $s = 2$, so the total effective pump is $4P_{\mathrm{th}}$.

For $J = 0$, if both components have nonzero steady-state densities, gain-loss balance gives

$$n^*_{A,+} = n^*_{A,-} = \frac{\gamma_C}{R}.$$

Writing

$$D_I = \frac{\gamma_I + W}{\gamma_I + W + 2\Gamma_{s,I}},$$

we obtain

$$n^*_{I,\pm} = \frac{sP_{\rm th}}{W}(1 \pm \varepsilon D_I),$$
$$n^*_{\pm} = \frac{\gamma_A}{R}\left[s(1 \pm \varepsilon D_I) - 1\right]. \tag{6}$$

These expressions apply when both predicted densities are positive:

$$s(1 - |\varepsilon| D_I) > 1.$$

For the chosen parameters, $D_I = 0.34180$, and this condition holds throughout the interval $-1 \le \varepsilon \le 1$ considered here.

The normalized condensate spin imbalance, or degree of circular polarization, is

$$z = \frac{n_+ - n_-}{n_+ + n_-} = \frac{s\varepsilon D_I}{s - 1}. \tag{7}$$

For $s = 2$, this gives $z = 0.68360\varepsilon$. Spin relaxation in the inactive reservoir reduces the difference between the particle transfer rates feeding the two components. The active reservoirs have equal steady-state densities, while the unequal transfer rates produce different condensate densities.

Numerical integration reproduces Eq. (6). For example, at $\varepsilon = 0.3$,

$$n^*_+ = 2.86924, \qquad n^*_- = 1.89267\ \mu\mathrm{m}^{-2}, \qquad z = 0.20508.$$

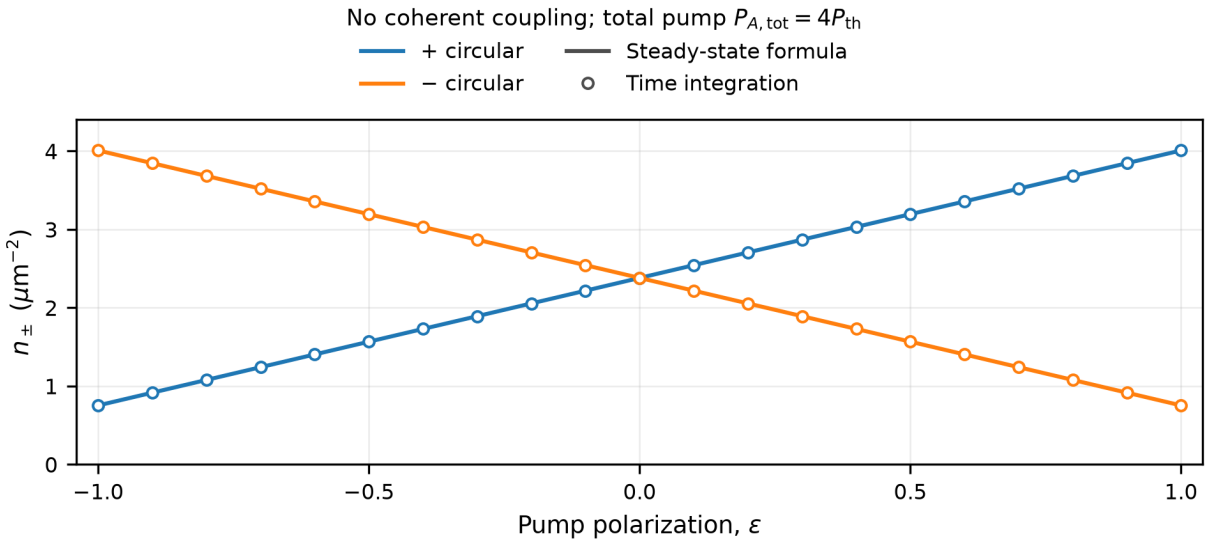


Figure 2: Steady-state densities of the circularly polarized components n_+ and n_- versus pump imbalance ε. The total effective pump is fixed at $P_{A,+} + P_{A,-} = 4P_{\rm th}$, and coherent coupling is absent, $J = 0$. Lines show the analytical predictions of Eq. (6); points show independent time integrations at 21 values of ε. Numerical values are evaluated over 1500–2000 ps. All runs use the same initial component densities and empty reservoirs.

2.3 Spatial distributions under ring-shaped pumping

For the spatially resolved problem, we use a ring-shaped pump

$$P_{A,\pm}(r) = s_{\mathrm{ring}} P_{\mathrm{th}}(1 \pm \varepsilon) \exp\left[-\frac{(r - r_0)^2}{d^2}\right], \tag{8}$$

with $r_0 = 10$ μm, $d = 2.5$ μm, and $s_{\mathrm{ring}} = 3.16875$. We compare $\varepsilon = 0$ and 0.3 at $J = 0$.

For equal pumping, the pump amplitude for each component equals that used in the scalar ring calculation of the previous report. The total peak pump is $6.3375 P_{\mathrm{th}}$, identical in both spinor calculations.

The initial fields are chosen as

$$\psi_+(\mathbf{r}, 0) = \psi_-(\mathbf{r}, 0) = \frac{\psi_{\mathrm{seed}}(\mathbf{r})}{\sqrt{2}},$$

where ψ_{seed} is the previous spatial seed consisting of 24 plane waves with an envelope in the pumped region. All reservoirs are initially empty.

The effect of pump imbalance is assessed using density maps with a common absolute color scale and the integrated quantities

$$N_\pm = \int n_\pm(\mathbf{r})\, d^2 r, \qquad Z = \frac{N_+ - N_-}{N_+ + N_-}.$$

The homogeneous expression (7) cannot be applied directly here, because spatial transport and reservoir inhomogeneity are included explicitly.

The maps in Fig. 3 correspond to the common final time $t = 2300$ ps. Populations are integrated over the full 256×256 μm domain.

ε	N_+	N_-	Z	Stationarity
0	305.8816	305.8816	0.00000	yes
0.3	440.2656	169.6108	0.44379	yes

Both condensate components and their reservoirs pass the stationarity criteria over two consecutive 100 ps windows. For symmetric pumping and equal initial fields, the component densities coincide. This checks the symmetric preparation; it does not establish convergence to equal densities from arbitrary initial conditions. At $\varepsilon = 0.3$, the pump imbalance mainly changes the component populations, giving $Z = 0.44379$. The spatial profiles remain similar, but do not coincide. Their relative difference after normalization by their respective populations is defined as

$$\widehat{n}_\sigma = \frac{n_\sigma}{N_\sigma}, \qquad \|f\|_{2,h} = \left(\Delta x\, \Delta y \sum_{i,j} |f_{ij}|^2\right)^{1/2},$$

$$\delta_{\mathrm{shape}} = \frac{\|\widehat{n}_+ - \widehat{n}_-\|_{2,h}}{\|\widehat{n}_+\|_{2,h}} = 5.55\%.$$

The norm uses all grid nodes in the 256×256 μm domain, without translation or rotation. The denominator is the norm of the normalized + component profile. The maps in Fig. 3 show absolute densities; normalization is used only to compare profile shapes.

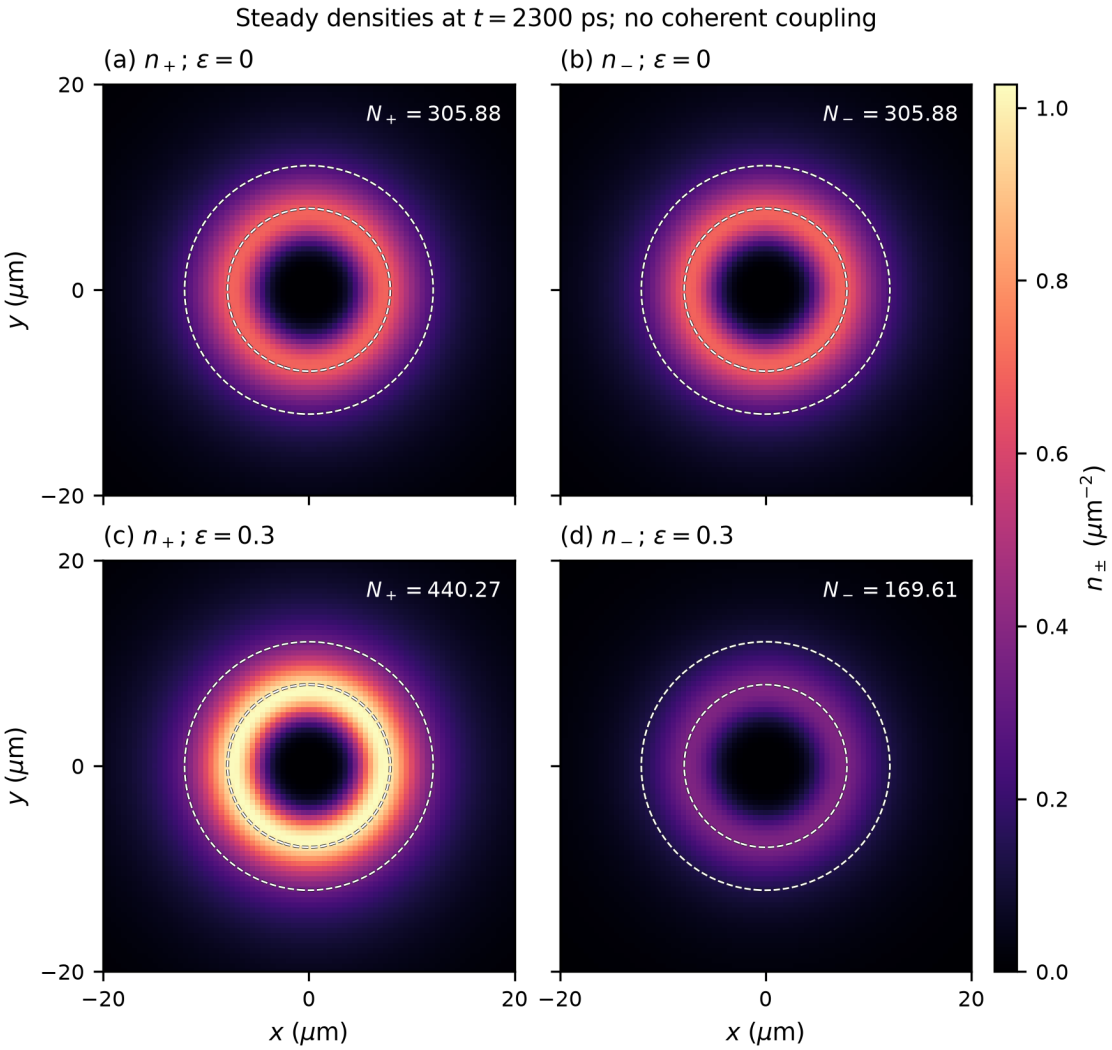


Figure 3: Spatial densities of the circularly polarized components under ring-shaped pumping at $J = 0$ and $t = 2300$ ps. The upper row corresponds to $\varepsilon = 0$, and the lower row to $\varepsilon = 0.3$; the left and right columns show $n_+(x, y)$ and $n_-(x, y)$, respectively. All maps use a common absolute color scale in μm^{-2}. Dashed lines mark the half-maximum pump contours. The central region $|x|, |y| \leq 20$ μm is shown; the populations $N_\pm$ indicated on the maps and the polarization Z reported in the text are evaluated over the full computational domain. Ring parameters: $r_0 = 10$ μm, $d = 2.5$ μm, and $s_{\mathrm{ring}} = 3.16875$. The integrated populations and stationarity status are given in the text.

3 Internal Josephson dynamics in the 0D model

In this section, we consider the spatially homogeneous (0D) limit of Eq. (4). The condensate fields, reservoir densities, and pump rates have no spatial dependence, so the kinetic terms vanish. Figures 4–6 are obtained from separate homogeneous simulations and do not describe the spatial dynamics of the ring-shaped condensate in Fig. 3.

3.1 Coherent coupling and particle exchange

In this section, the term $-J\psi_{-\sigma}$ is retained in Eq. (4). It describes coherent coupling between the two circularly polarized components of a single condensate. For equal diagonal energies, the corresponding splitting between the two linearly polarized eigenstates is $2J$.

Here, J is independent of the wave vector. The internal Josephson dynamics considered here therefore do not require momentum-dependent TE–TM splitting. Incoherent pumping supplies particles through the reservoirs, while the relative phase emerges through the dynamics of the coupled fields. Internal Josephson coupling under nonresonant pumping is studied, for example, in Ref. [4].

The two circularly polarized components are written as

$$\psi_{\pm}(t) = \sqrt{n_{\pm}(t)}\, e^{i\phi_{\pm}(t)}.$$

Their relative phase is

$$\theta(t) = \phi_{+}(t) - \phi_{-}(t).$$

Here, a fixed relative phase means that θ is constant in time. The individual phases may continue to evolve, but at the same rate. Their difference need not be zero. Spatial uniformity is assumed in the 0D model; it is not a result of the phase-locking calculation.

The Josephson current between the components is

$$I_J = \frac{2J}{\hbar}\,\mathrm{Im}(\psi_{-}^{*}\psi_{+}) = \frac{2J}{\hbar}\sqrt{n_{+}n_{-}}\,\sin\theta. \tag{9}$$

With this convention, positive I_J denotes particle transfer into the + component:

$$\begin{aligned} \dot{n}_{+} &= (Rn_{A,+} - \gamma_C)n_{+} + I_J, \\ \dot{n}_{-} &= (Rn_{A,-} - \gamma_C)n_{-} - I_J. \end{aligned} \tag{10}$$

The relative phase obeys

$$\dot{\theta} = -\frac{V_{+} - V_{-}}{\hbar} - \frac{2J}{\hbar}\frac{z}{\sqrt{1-z^2}}\cos\theta. \tag{11}$$

The difference in energy shifts drives the relative phase evolution. The coupling J affects the phase and transfers particles between the components, changing both their densities and their energy shifts.

3.2 Three observed dynamical regimes

The calculations use $s = 2$, $\varepsilon = 0.3$, and identical initial conditions. Three coupling strengths are considered:

- $J = 5$ μeV: the densities oscillate periodically, and the relative phase continues to wind;
- $J = 6.5$ μeV: density oscillations persist, but the relative phase oscillates within a bounded interval;
- $J = 10$ μeV: the densities and relative phase become constant in time.

Thus, vanishing mean relative-phase drift does not imply that density oscillations have ceased.

At $J = 10$ μeV, the steady Josephson current is

$$I_J \simeq 0.01477\ \mu\mathrm{m}^{-2}\,\mathrm{ps}^{-1}.$$

It compensates the unequal gain-loss balance in the two components. A fixed relative phase can therefore coexist with a nonzero Josephson current. The complex fields themselves continue to rotate at a common frequency.

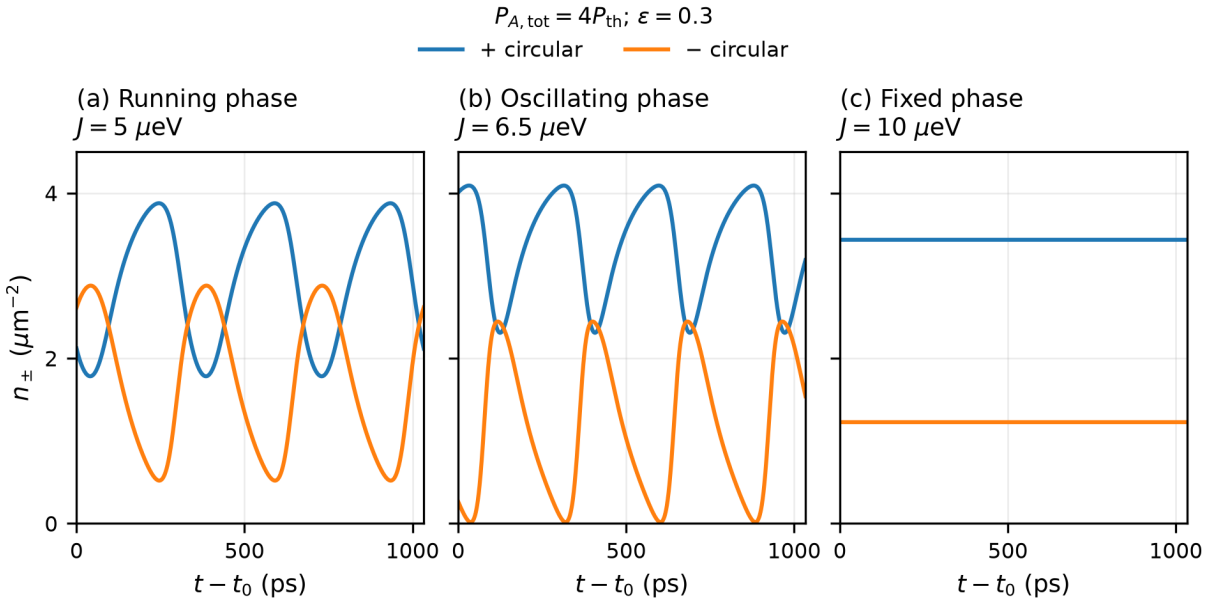


Figure 4: Spatially homogeneous condensate densities $n_+(t)$ and $n_-(t)$ at three coherent coupling strengths: **(a)** $J = 5$, **(b)** 6.5, and **(c)** 10 μeV. In all cases, $s = 2$ and $\varepsilon = 0.3$, with identical initial conditions. Late-time intervals are shown after verification of the long-time dynamical regime. The horizontal axis is $\tau = t - t_0$; all panels use $t_0 = 16966.5$ ps and end at 18000 ps. The displayed intervals have equal durations and share the same density scale.

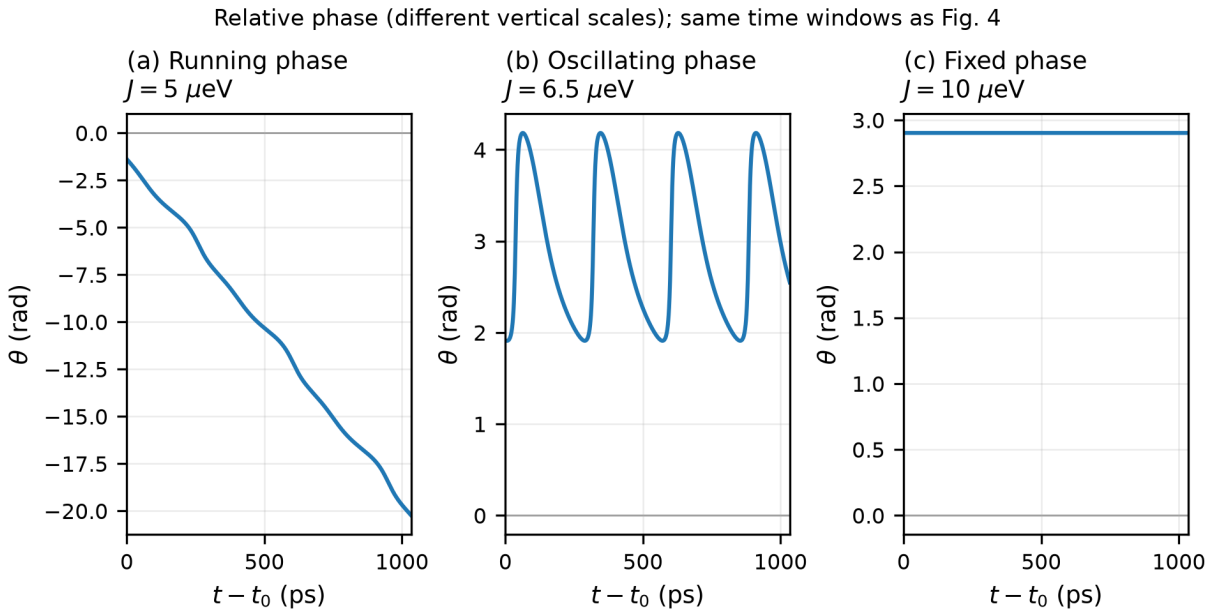


Figure 5: Relative phase $\theta(t)$ between the two circularly polarized components in the homogeneous model, for the same trajectories and time intervals as in Fig. 4. The panels correspond to $J = 5,\ 6.5,\ 10\ \mu$eV. Weak coupling produces a running relative phase; at intermediate coupling the phase oscillates within a bounded interval; at stronger coupling it becomes constant in time. The phase is continuously unwrapped. Only shifts by integer multiples of 2π are used to position the curves, preserving the phase modulo 2π.

3.3 Transition to steady densities and a fixed relative phase

Two quantities characterize the dependence on J:

$$A_z = \frac{\max z - \min z}{2}, \qquad \Delta\overline{\omega} = \overline{\omega_+ - \omega_-} = -\overline{\dot{\theta}}. \tag{12}$$

The first measures the amplitude of spin-imbalance oscillations. The second is the mean frequency difference between the components, using the convention $\omega_\sigma = -\arg\dot{\psi}_\sigma$. For periodic motion with period T, it is evaluated as

$$\Delta\overline{\omega} = -\left\langle \frac{\theta(t+T) - \theta(t)}{T} \right\rangle_t .$$

The average is taken over all admissible t within the validated late-time window. The minus sign converts the mean phase drift into the frequency difference. For steady and uncoupled components, the negative slope of the relative phase is used.

In the calculated series, the mean phase drift vanishes near $J = 6.4\ \mu$eV, whereas density oscillations persist up to stronger coupling. The transition to steady densities and a fixed relative phase lies between the sampled points $J = 6.84$ and $6.86\ \mu$eV.

At $J = 6.38$ and $6.40\ \mu$eV, one component becomes too weak during part of the cycle for its phase to be determined reliably. Density periodicity is confirmed at these points, but no value of $\Delta\overline{\omega}$ is assigned to them.

This boundary separates dynamical regimes of an already existing condensate. Its location depends on the pump, model coefficients, and preparation protocol. Each point here is obtained from an independent run with the same initial state; the resulting dependence is not a complete map of all possible states.

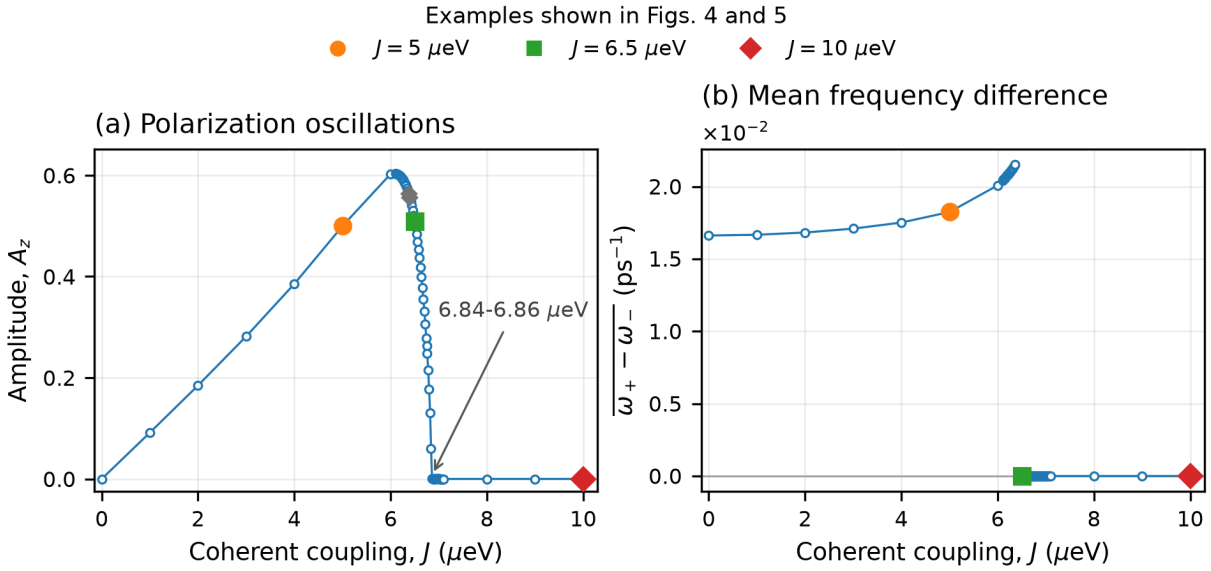


Figure 6: Change of dynamical regime in the homogeneous model with increasing coupling J, at $s = 2$, $\varepsilon = 0.3$. **(a)** Spin-imbalance oscillation amplitude A_z. **(b)** Mean frequency difference $\Delta\overline{\omega}$, in ps^{-1}. Each point is an independent start with identical initial conditions after transient validation. Frequencies use the negative phase change per full period. Gray diamonds at $J = 6.38$ and 6.40 μeV indicate cycles with an unreliable phase due to low component density; frequency gaps do not mean zero. Colored markers denote $J = 5,\ 6.5,\ 10$ μeV, used in Figs. 4 and 5. The arrow labels the sampled interval 6.84–6.86 μeV where the first density cycle ceases.

A Parameter choice and matching

A.1 Parameters of the previous model

The base coefficients are retained from the previous report, which used the model of Alyatkin et al. [1]. In particular, R is a model parameter, and $g_R = 2g_C$ is an assumption of that model. Retaining these values makes the comparison consistent; it does not imply that all coefficients have been independently measured for the structure considered here.

Quantity	Value
$\hbar$	0.658212 meV ps
m/m_e	5.63×10^{-5}
$K = \hbar^2/(2m)$	0.676729 meV μm^2
$g_C = \alpha_1$	0.0024 meV μm^2
$g_R = g_A$	0.0048 meV μm^2
R	0.021 μm^2 ps^{-1}
γ_C	0.182 ps^{-1}
$\gamma_R = \gamma_A$	0.050 ps^{-1}
η	9.8 ps

All densities n are expressed in μm^{-2}, and pump rates in μm^{-2} ps^{-1}.

A.2 Inactive reservoir

The matching condition $g_I/W = g_R\eta$ fixes the ratio of the two new coefficients, but not their individual values. The main parameter set additionally assumes $g_I = g_R$. Hence,

$$W = \frac{1}{\eta} = 0.102041 \text{ ps}^{-1}.$$

For the intrinsic decay of the inactive reservoir, we use $\gamma_I = 1/550$ ps^{-1}, following the scale in the model of Gnusov et al. [2]. The remaining parameters of that work are not adopted as a complete set: γ_A is retained from the previous report, while W is fixed by the matching condition above.

The resulting values are

$$g_I = 0.0048 \text{ meV } \mu\text{m}^2, \qquad \tau_I = \frac{1}{\gamma_I + W} = 9.62844 \text{ ps},$$

$$P_{\text{in}} = 1.01782 P_A.$$

The time scale of 550 ps describes particle loss without transfer to the active reservoir. The response of the inactive reservoir is governed by the sum of the decay and transfer rates, giving approximately 9.63 ps.

A.3 Spin-dependent coefficients

For interactions between polaritons with the same circular polarization, we retain $\alpha_1 = g_C$. For opposite circular polarizations, we choose

$$\alpha_2 = -0.1\alpha_1 = -0.00024 \text{ meV}\,\mu\text{m}^2.$$

The ratio -0.1 corresponds to the model choice in Askitopoulos et al. [4]. The reservoir spin-relaxation rates are set to

$$\Gamma_{s,A} = \Gamma_{s,I} = 0.1 \text{ ps}^{-1},$$

following the scale used by Gnusov et al. [3]. These are fixed parameters of the extension, rather than measurements of the present structure.

When the spin-flip term is written as $\Gamma_s(n_{-\sigma} - n_\sigma)$, spin exchange alone causes the population difference to decay at a rate $2\Gamma_s$. The corresponding relaxation time is 5 ps.

Equation (4) reduces exactly to the scalar two-reservoir model, Eq. (2), when $J = \Gamma_{s,A} = \Gamma_{s,I} = 0$ and the pump, condensate field, and both reservoir densities of the $-$ component vanish. The zero fields and reservoir densities are imposed initially and then remain zero. The matching conditions (3) preserve the stationary solutions of the earlier one-reservoir model, not its arbitrary transient dynamics. For the main parameter set, nonzero Γ_s transfers particles between reservoir spin components.

A.4 Coherent coupling scale

The reference value $J = 0.010$ meV corresponds to a linear polarization splitting of $2J = 20$ μeV. Splittings of approximately 15–20 μeV were measured by Paraïso et al. [5] in a different microcavity structure. That result is used only as a reference energy scale; the dependence on J is investigated separately.

The additional coefficients thus form a consistent phenomenological extension of the previous model. The calculations are not claimed to reproduce a particular experiment quantitatively.

B Numerical method and validation

In the homogeneous calculations, spatial derivatives are omitted, but the units of two-dimensional density are retained. The fields and reservoirs are integrated jointly with DOP853, using relative and absolute tolerances of 2×10^{-11} and 2×10^{-13}, respectively. The spinor series starts from

$$\psi_{\pm}(0) = \sqrt{5 \times 10^{-5}}\, e^{\pm 0.1i}, \qquad n_{A,\pm}(0) = n_{I,\pm}(0) = 0.$$

Steady-state values at $J = 0$ are checked against Eq. (6). For oscillatory regimes, validation includes recurrence of the densities and polarization state, the period, and persistence of the amplitude when the simulation is extended. Selected points are also recomputed with tighter integration tolerances.

The spatial series uses a periodic domain of 256×256 μm, a 512×512 grid, and a time step $\Delta t = 0.025$ ps. Two complex fields and four real reservoir densities are stored; $n_{\pm}$ are evaluated as $|\psi_{\pm}|^2$.

Each time step consists of a kinetic half-step for both fields, a joint local fourth-order Runge–Kutta step, and a second kinetic half-step. The complete split-step scheme is second-order accurate in time. In Fourier space, the kinetic half-step multiplies each field by

$$\exp\left[-\frac{iK(k_x^2 + k_y^2)\Delta t}{2\hbar}\right].$$

All four spatial runs reached 2300 ps. Both final windows, 2100–2200 and 2200–2300 ps, pass the stationarity criteria: for each condensate and reservoir population, the range and half-window mean drift, relative to the window-mean magnitude, are below 0.5%; for each condensate and reservoir density map, the relative L2 change from the window start remains below 1%, checked every 5 ps. At $J = 0$, no relative-phase condition is imposed.

At $\varepsilon = 0.3$, independent runs use a halved time step and a coarser spatial grid. Relative errors are defined as

$$\delta N_\sigma = \frac{|N_\sigma^{(c)} - N_\sigma^{(\mathrm{ref})}|}{|N_\sigma^{(\mathrm{ref})}|},$$

$$\delta_2(n_\sigma) = \frac{\|n_\sigma^{(c)} - \mathcal{S} n_\sigma^{(\mathrm{ref})}\|_{2,h_c}}{\|\mathcal{S} n_\sigma^{(\mathrm{ref})}\|_{2,h_c}}.$$

Here ref is the main 512^2 run with a 0.025 ps step, and c is the check. The operator $\mathcal{S}$ samples the reference at check-grid nodes and is the identity for the time-step comparison. The norm covers the full domain on the check grid. The table lists $100\delta N_\sigma$ and $100\delta_2(n_\sigma)$ at the common final time, i.e. percent errors.

Check	δN_+	δN_-	$\delta_2(n_+)$	$\delta_2(n_-)$	Status
256^2 grid	9.67×10^{-7}	6.03×10^{-7}	3.27×10^{-5}	2.95×10^{-5}	PASS
0.0125 ps step	1.40×10^{-3}	1.76×10^{-3}	1.30×10^{-3}	1.68×10^{-3}	PASS

Both errors must be below 1% for each component. No rotation or amplitude fitting is applied. Maximum relative values over the four runs: total particle-balance residual 1.18×10^{-12}, boundary density fraction 3.05×10^{-7}, and high-wave-number spectral weight 2.59×10^{-10}. Their respective thresholds are 10^{-3}, 10^{-4}, and 10^{-6}. Overall assessment of the new spatial series: **PASS**.

References

[1] S. Alyatkin et al. Antiferromagnetic Ising model in a triangular vortex lattice of quantum fluids of light. *Science Advances* **10**(34), eadj1589 (2024). https://doi.org/10.1126/sciadv.adj1589

[2] I. Gnusov et al. Quantum vortex formation in the "rotating bucket" experiment with polariton condensates. *Science Advances* **9**(4), eadd1299 (2023). https://doi.org/10.1126/sciadv.add1299

[3] I. Gnusov et al. Optical orientation, polarization pinning, and depolarization dynamics in optically confined polariton condensates. *Physical Review B* **102**, 125419 (2020). https://doi.org/10.1103/PhysRevB.102.125419

[4] A. Askitopoulos et al. Nonresonant optical control of a spinor polariton condensate. *Physical Review B* **93**, 205307 (2016). https://doi.org/10.1103/PhysRevB.93.205307

[5] T. K. Paraïso et al. Multistability of a coherent spin ensemble in a semiconductor microcavity. *Nature Materials* **9**, 655–660 (2010). https://doi.org/10.1038/nmat2787